PRATICAL-10

AIM: Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

Steps:-

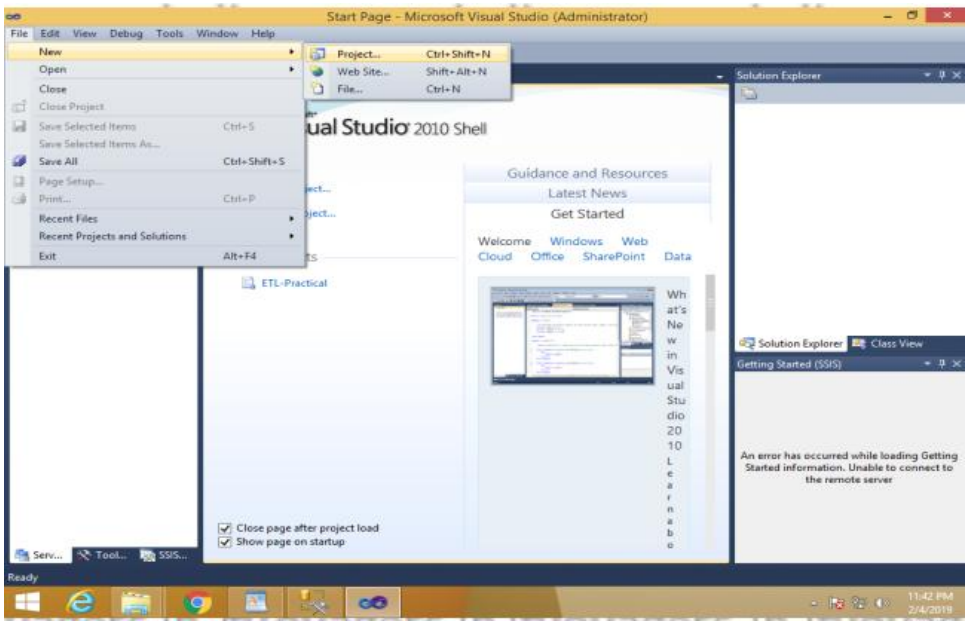
Step 1: Creating Data Warehouse

CREATED IN PRACTICAL 9

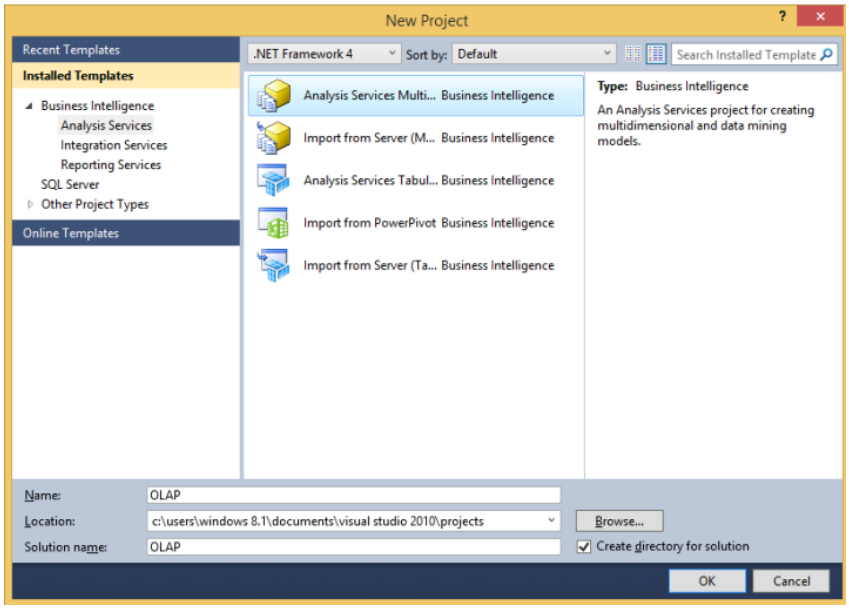
Step 2: Start SSDT environment and create New Data Source

Go to Sql Server Data Tools --> Right click and run as administrator

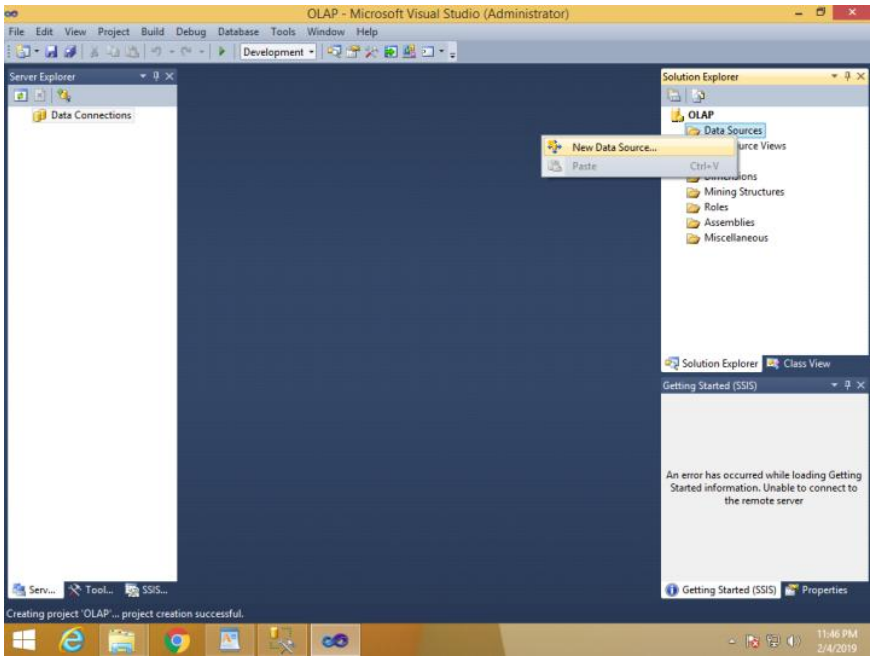
Click on File → New → Project



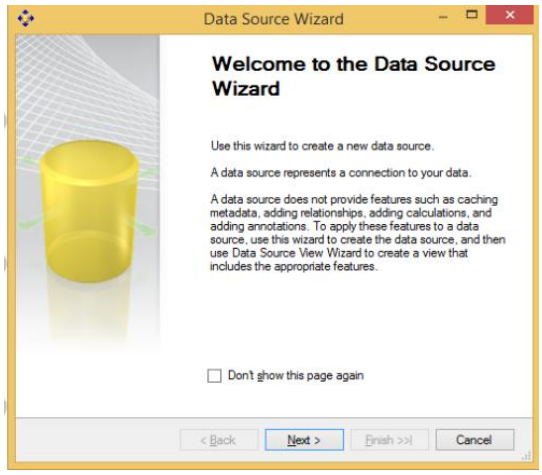
In Business Intelligence → Analysis Services Multidimensional and Data Mining models → appropriate project name → click OK



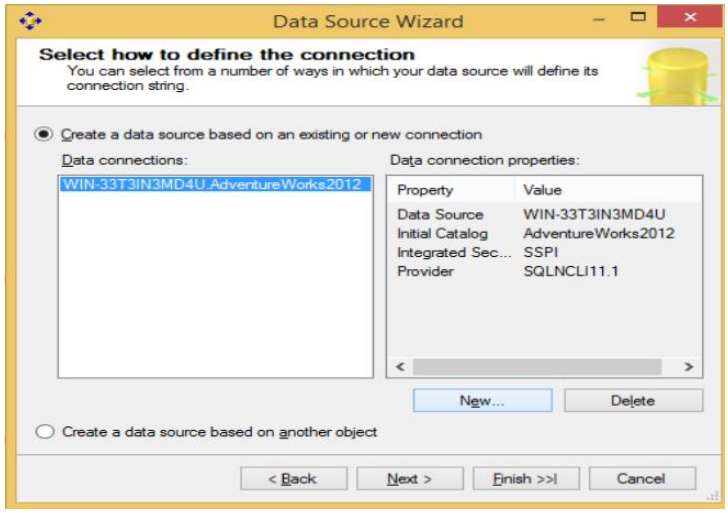
Right click on Data Sources in solution explorer → New Data Source



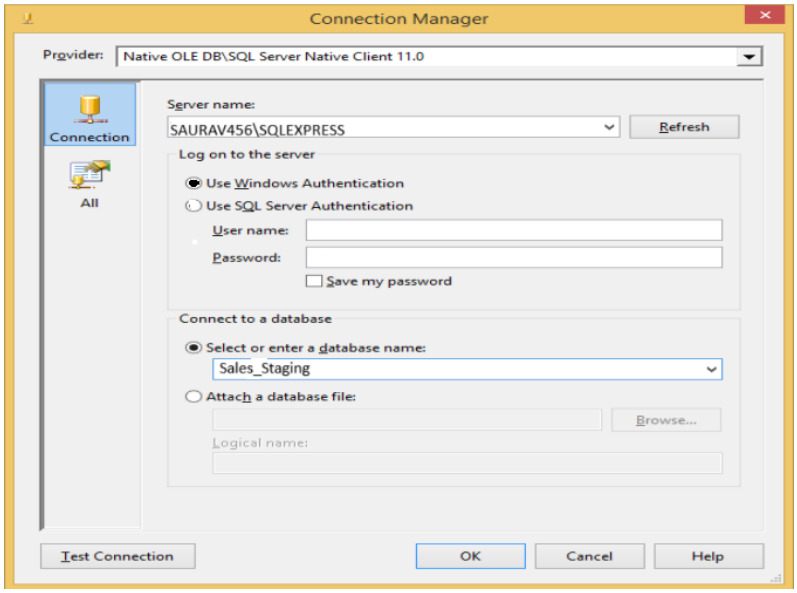
Data Source Wizard appears

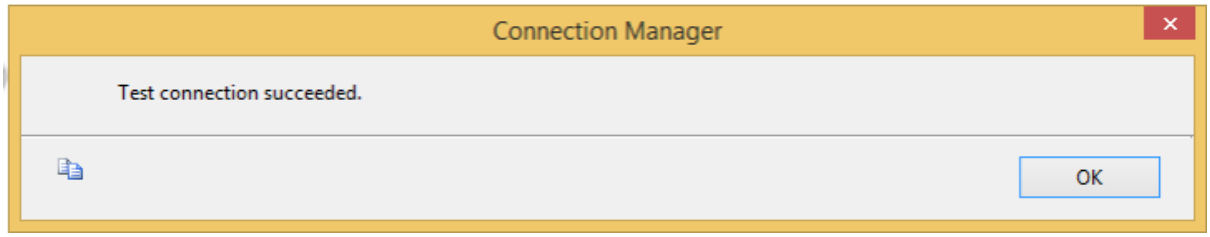


Click on New

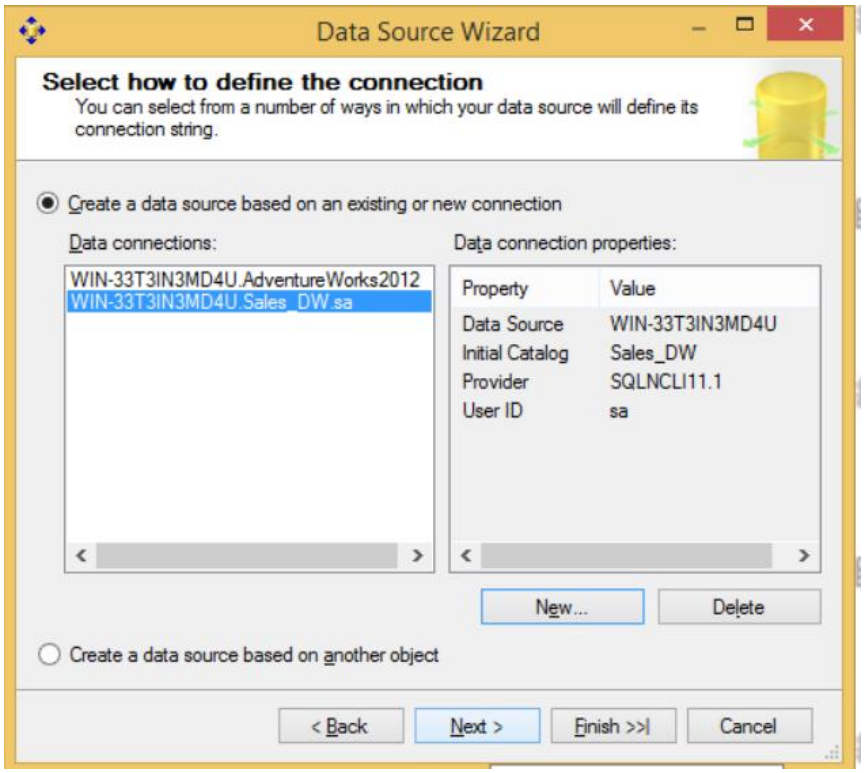


Select Server Name → select Use SQL Server Authentication → Select or enter a database name (Sales_Staging)

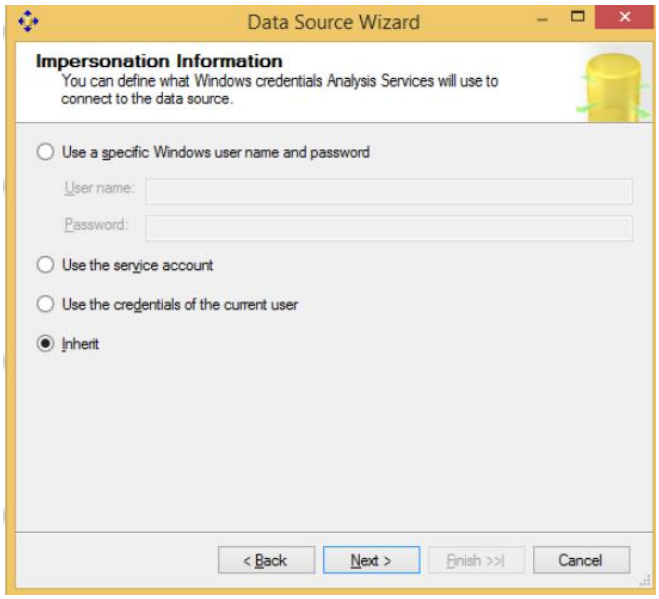




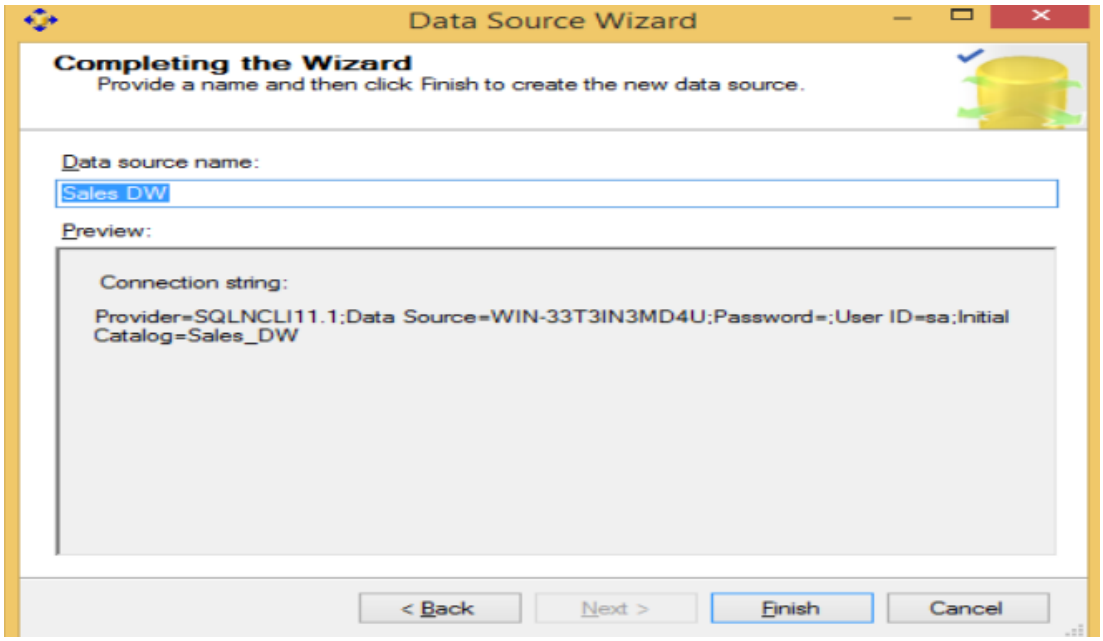
Click ok



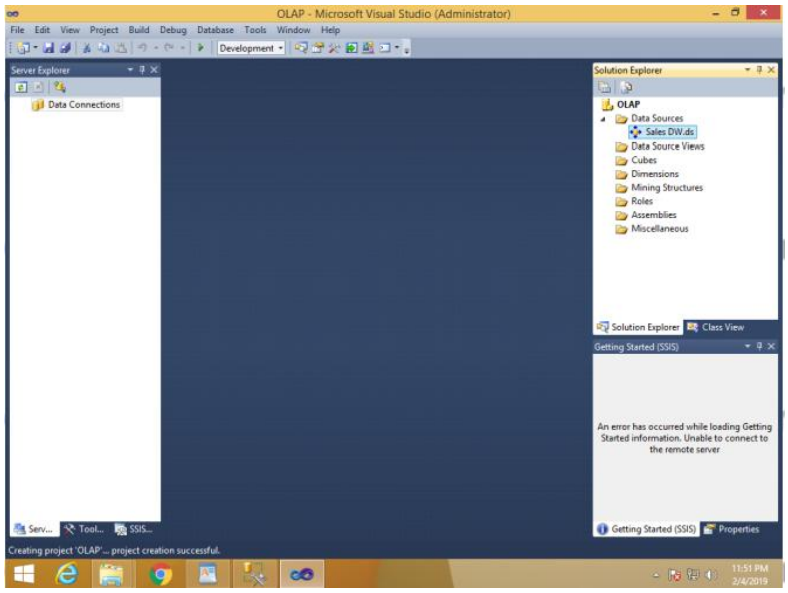
Select Inherit → Next



Click Finish

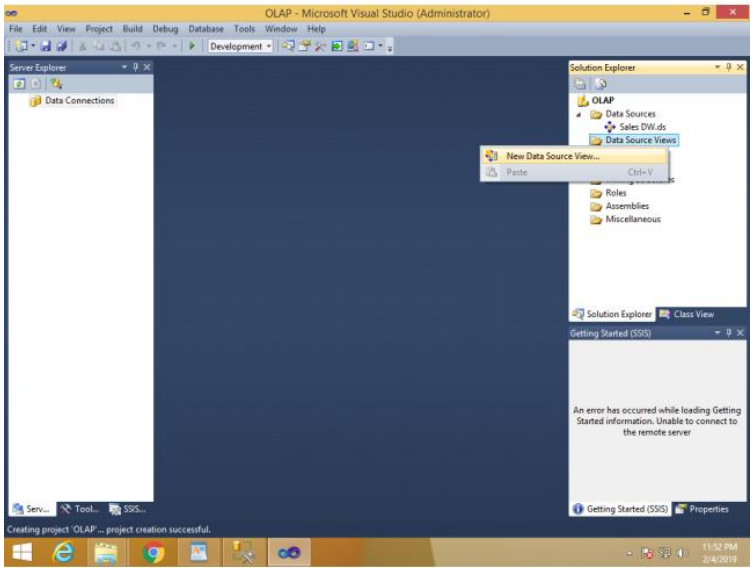


Sales_Staging.ds gets created under Data Sources in Solution Explorer

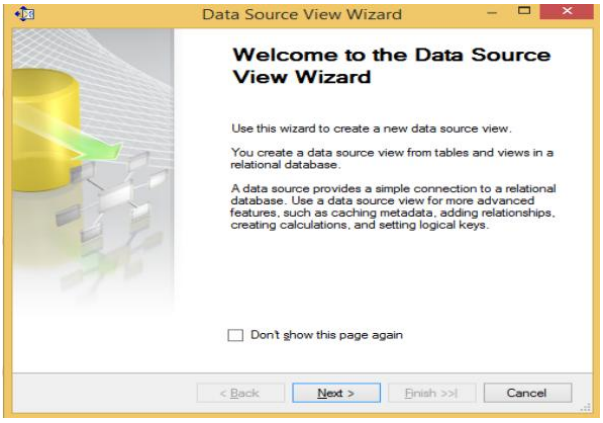


Step 3: Creating New Data Source View

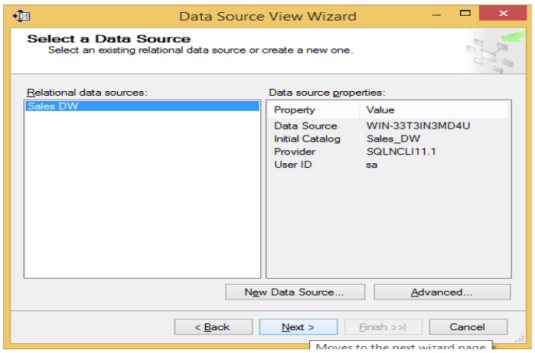
In Solution explorer right click on Data Source View → Select New Data Source View



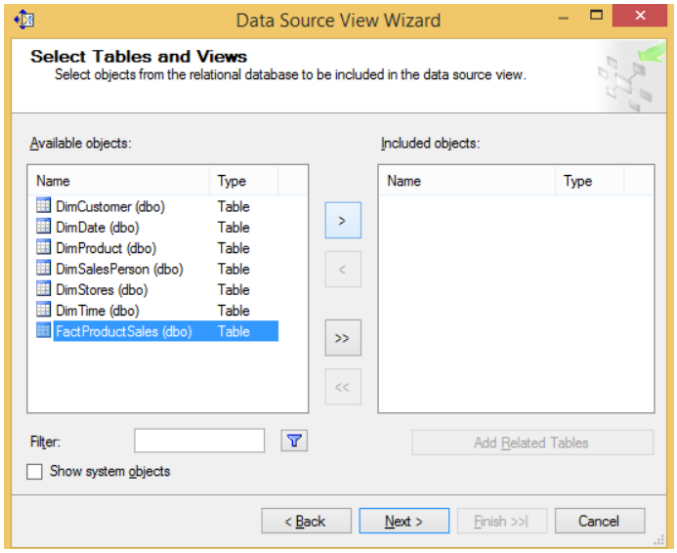
Click Next

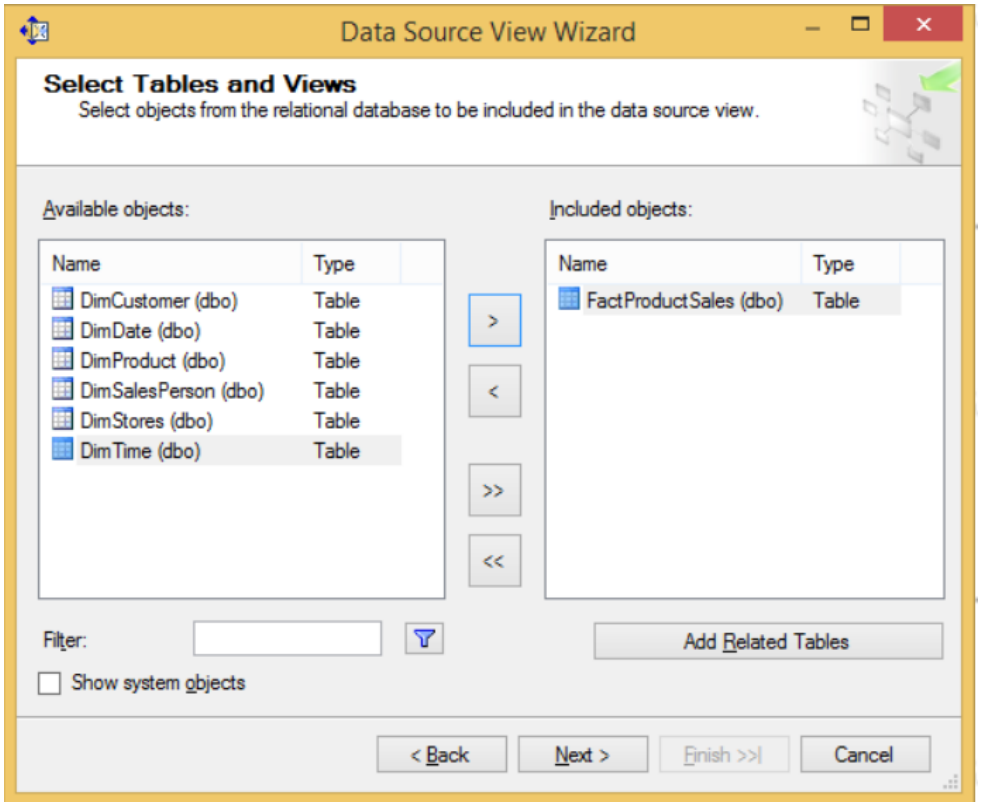


Click Next

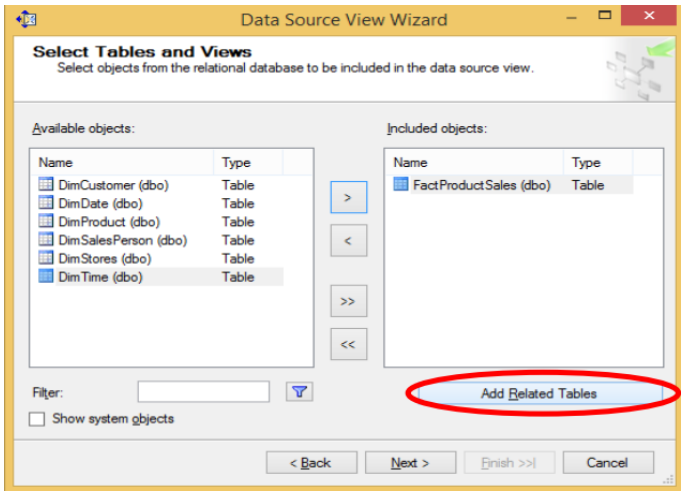


Select FactProductSales(dbo) from Available objects and put in Includes Objects by clicking on

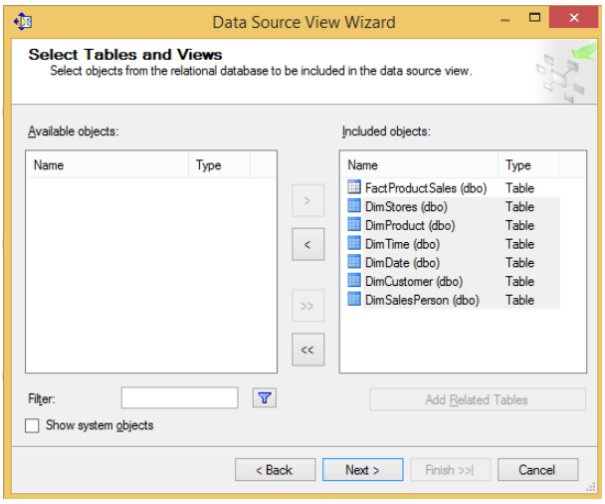




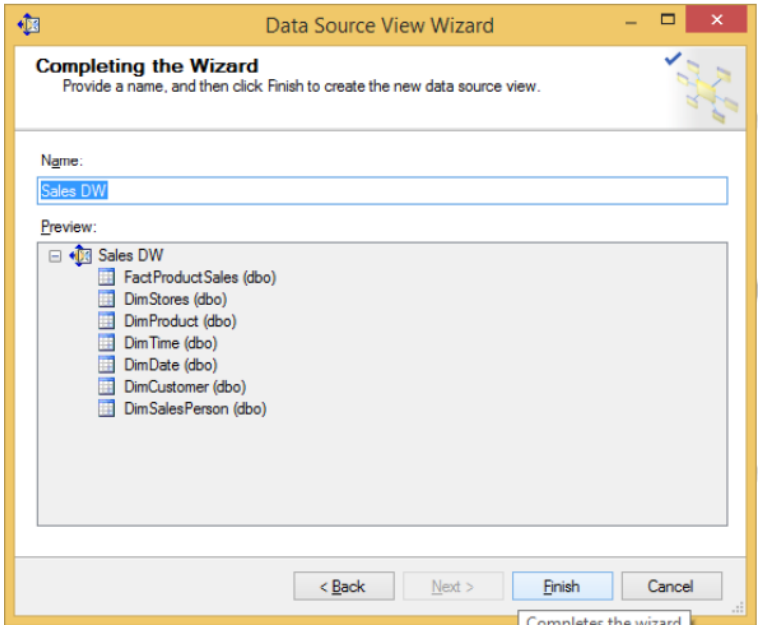
Click on Add Related Tables



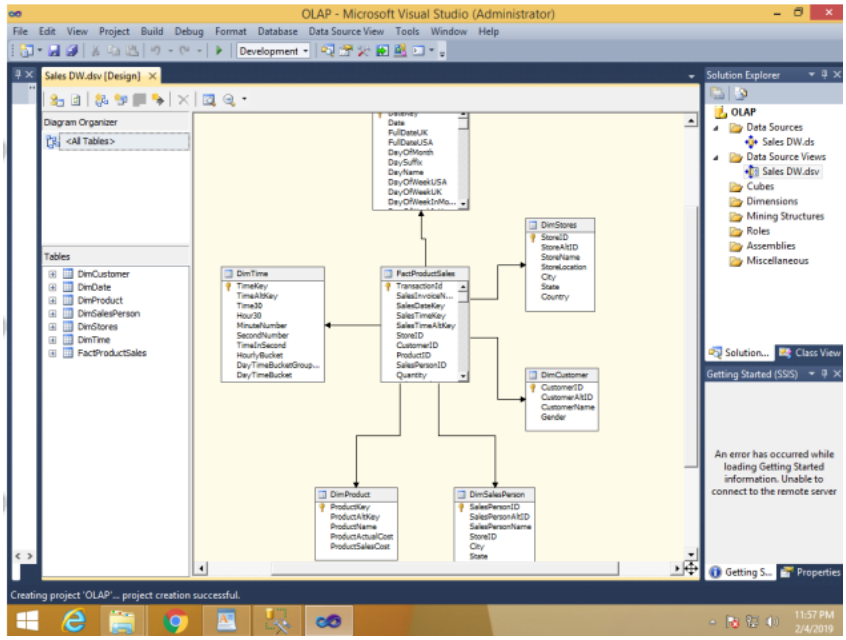
Click Next



Click Finish

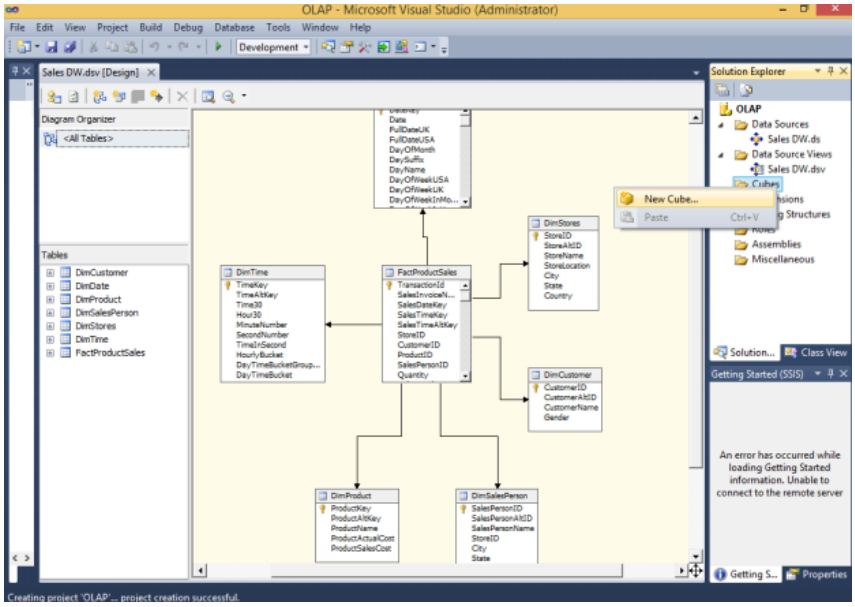


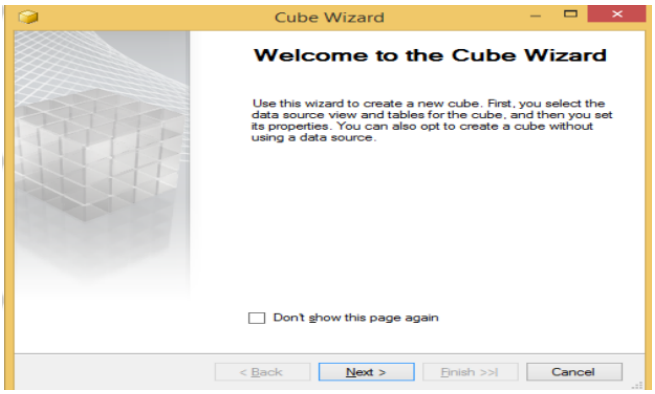
Sales_Staging.dsv appears in Data Source Views in Solution Explorer.



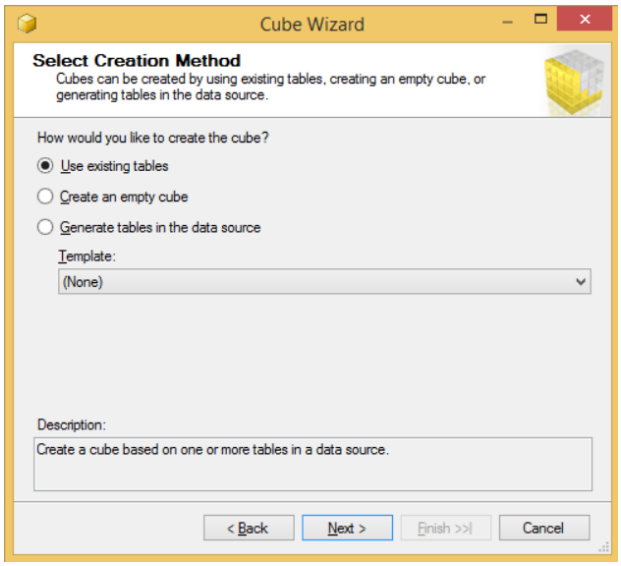
Step 4: Creating new cube

Right click on Cubes → New Cube

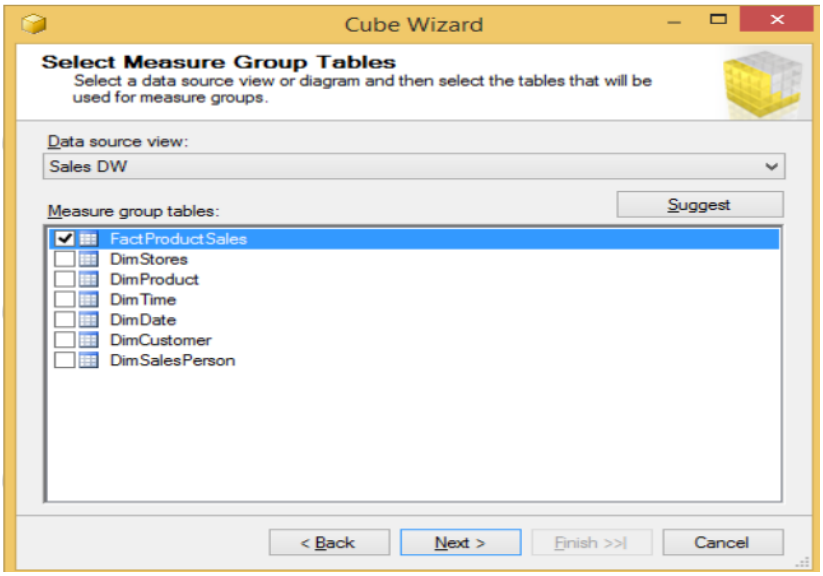




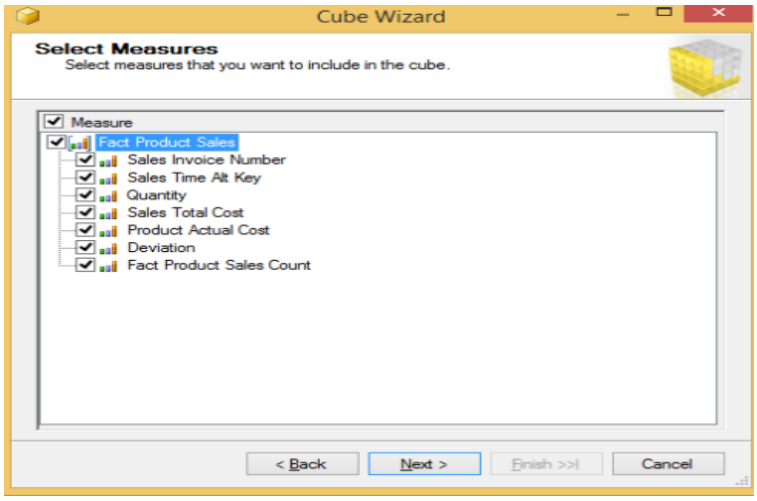
Select Use existing tables in Select Creation Method → Next



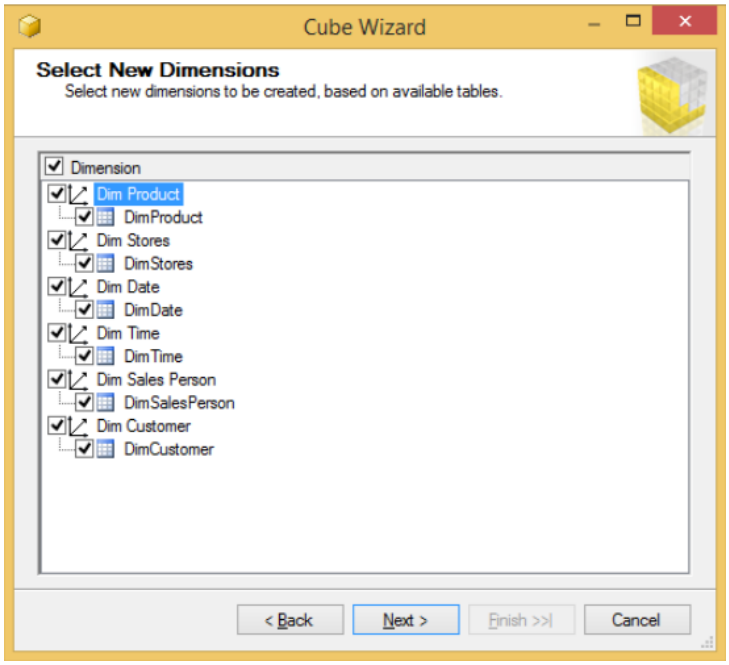
In Select Measure Group Tables → Select FactProductSales → Click Next



In Select Measures → check all measures → Next

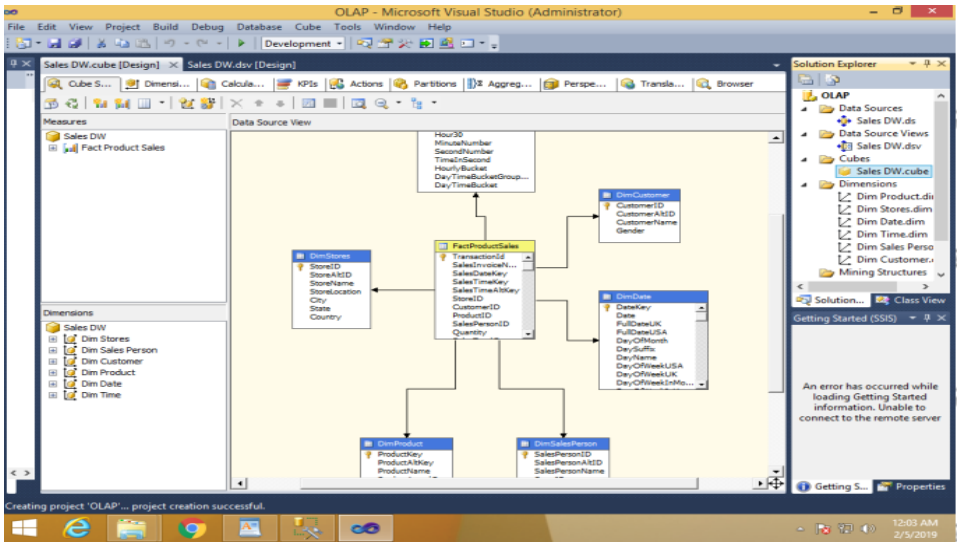


In Select New Dimensions → Check all Dimensions → Next



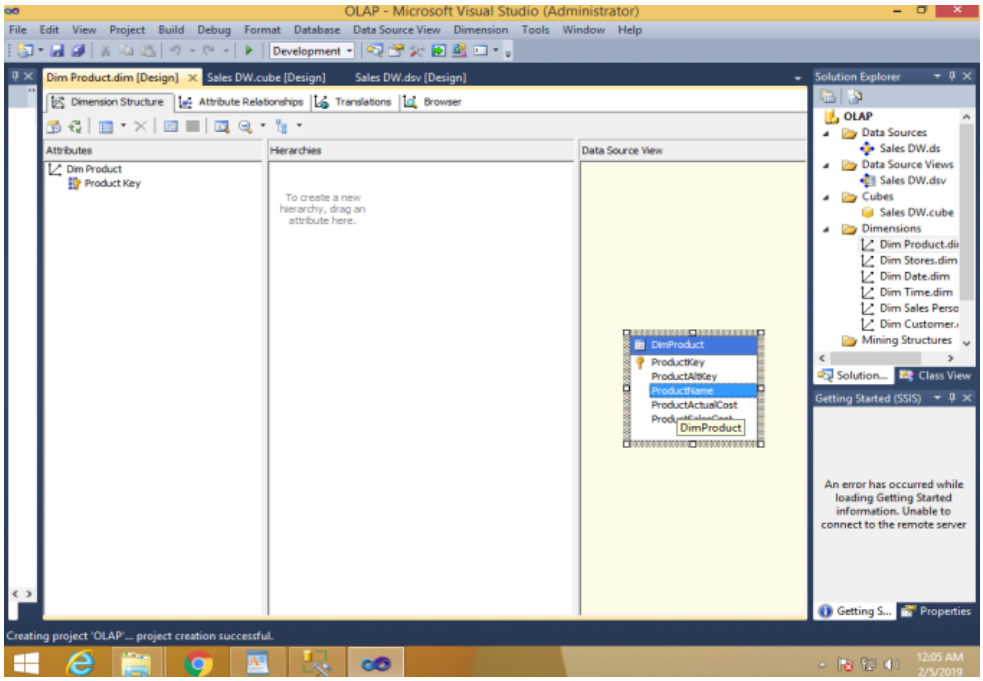
Click on Finish

Sales_Staging.cube is created

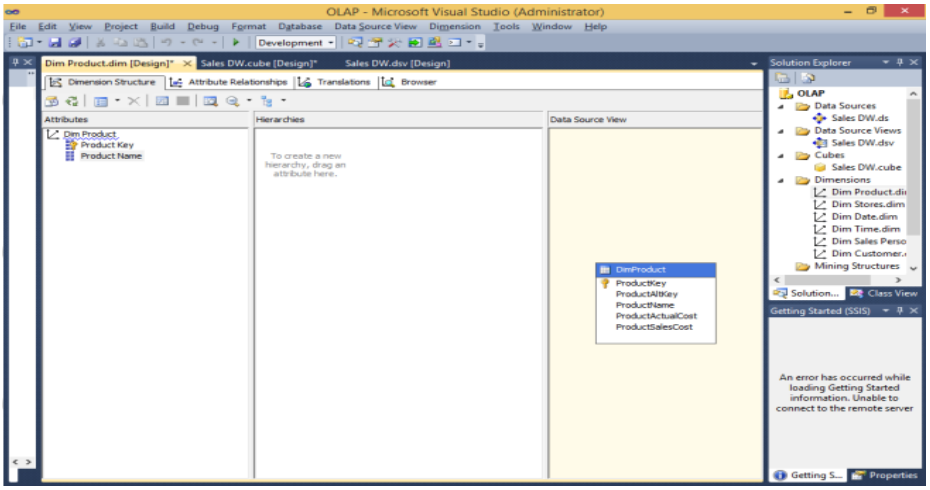


Step 5: Dimension Modification

In dimension tab → Double Click Dim Product.dim



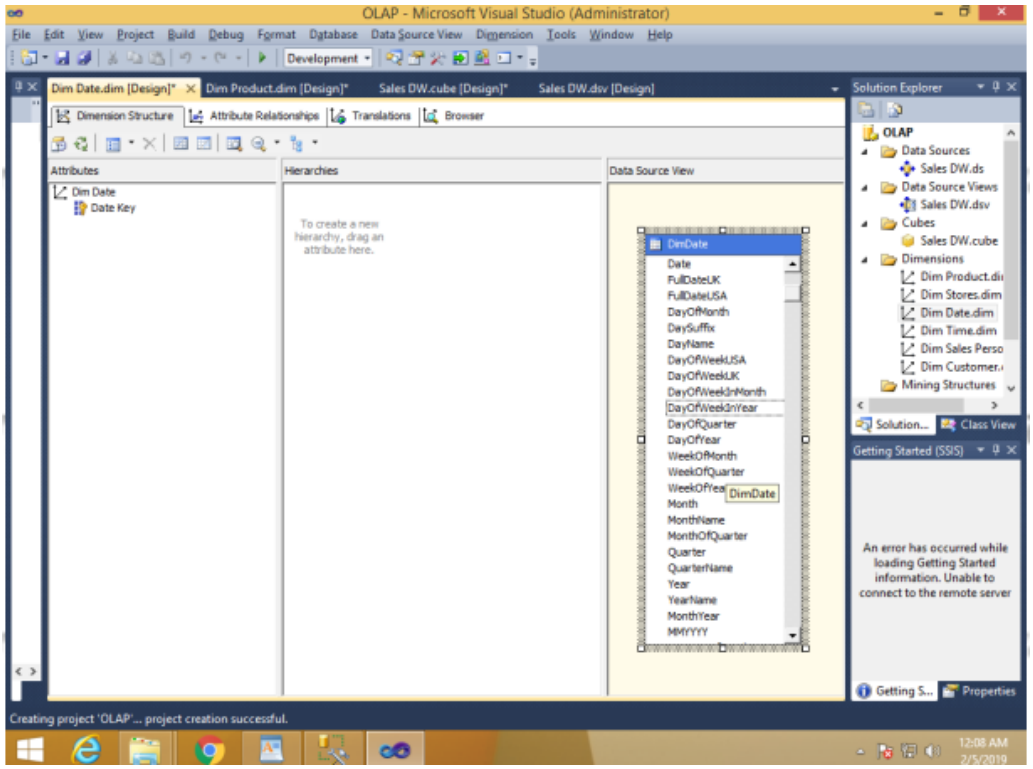
Drag and Drop Product Name from Table in Data Source View and Add in Attribute Pane at left side

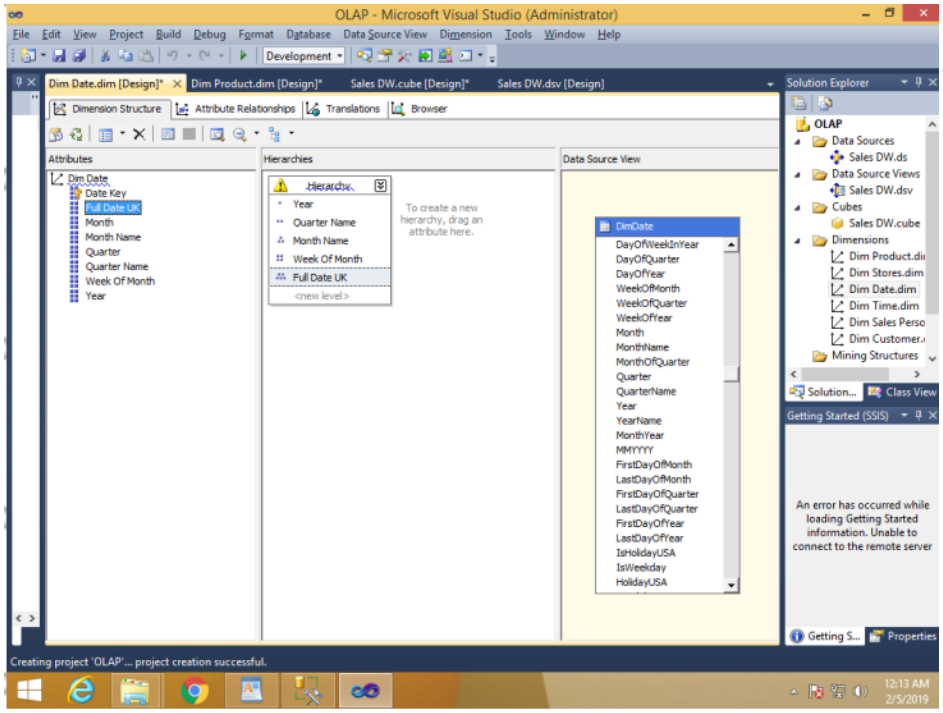


Step 6: Creating Attribute Hierarchy in Date Dimension

Double click On Dim Date dimension -> Drag and Drop Fields from Table shown in Data Source View to Attributes-> Drag and Drop attributes from leftmost pane of attributes to middle pane of Hierarchy.

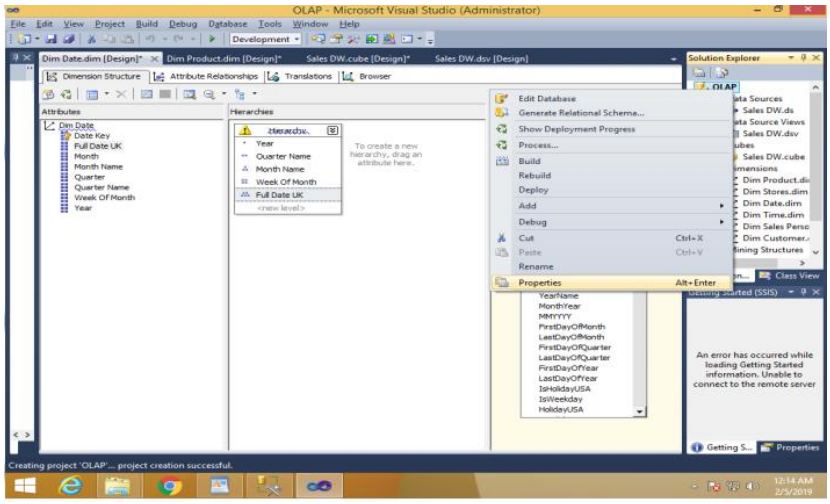
Drag fields in sequence from Attributes to Hierarchy window (Year, Quarter Name, Month Name, Week of the Month, Full Date UK)



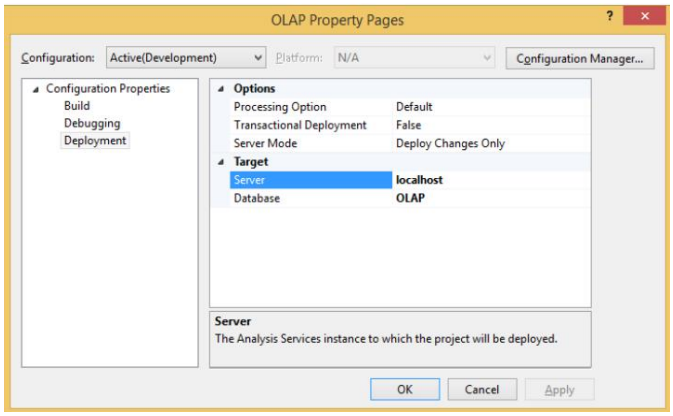


Step 7: Deploy Cube .

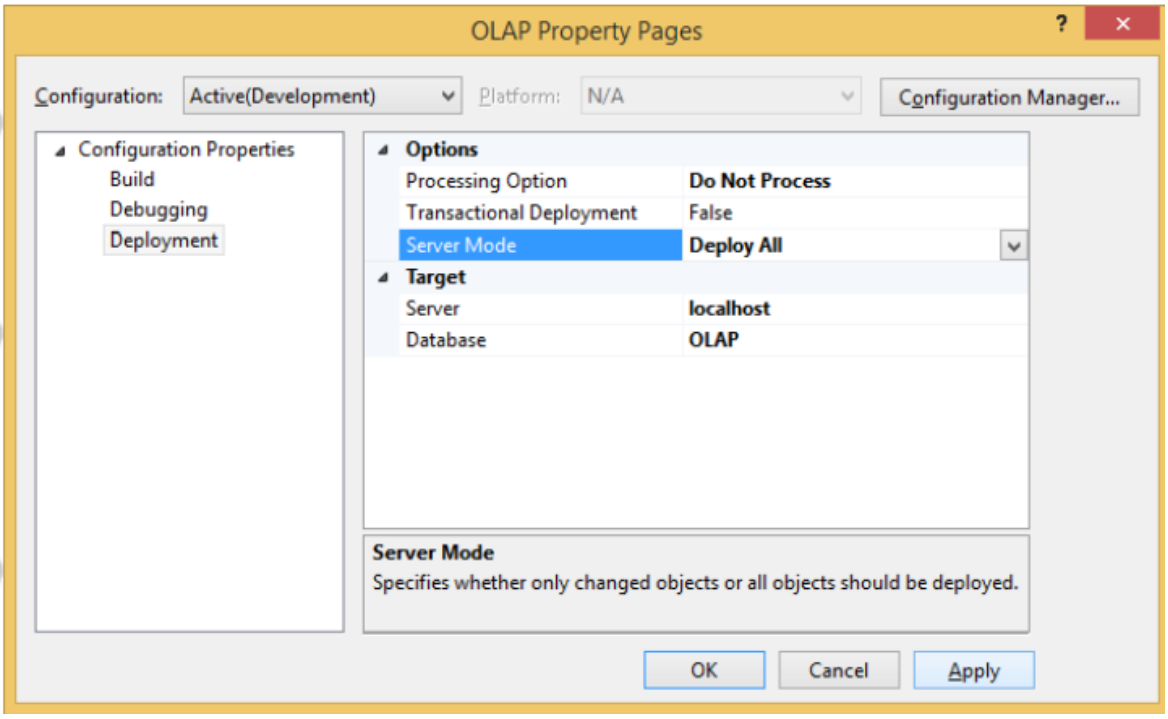
Right click on Project name → Properties



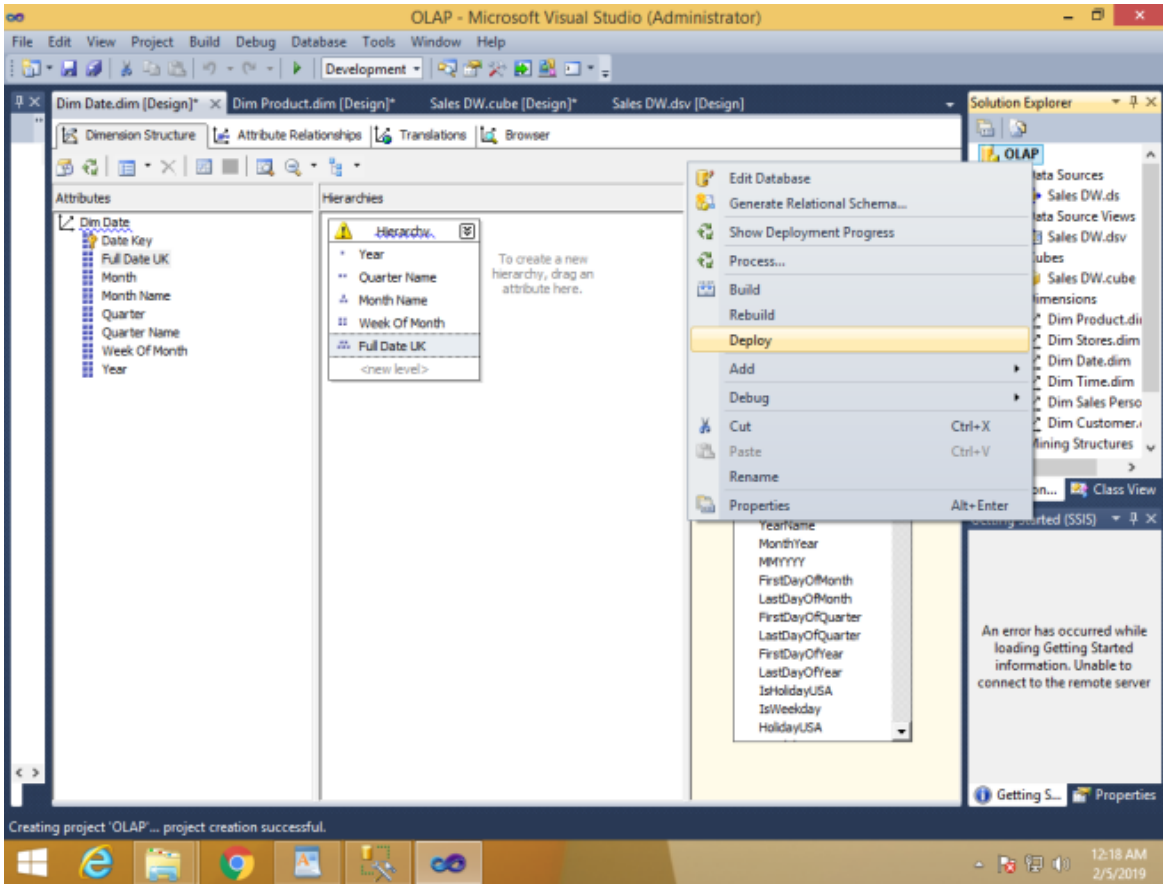
This window appaers



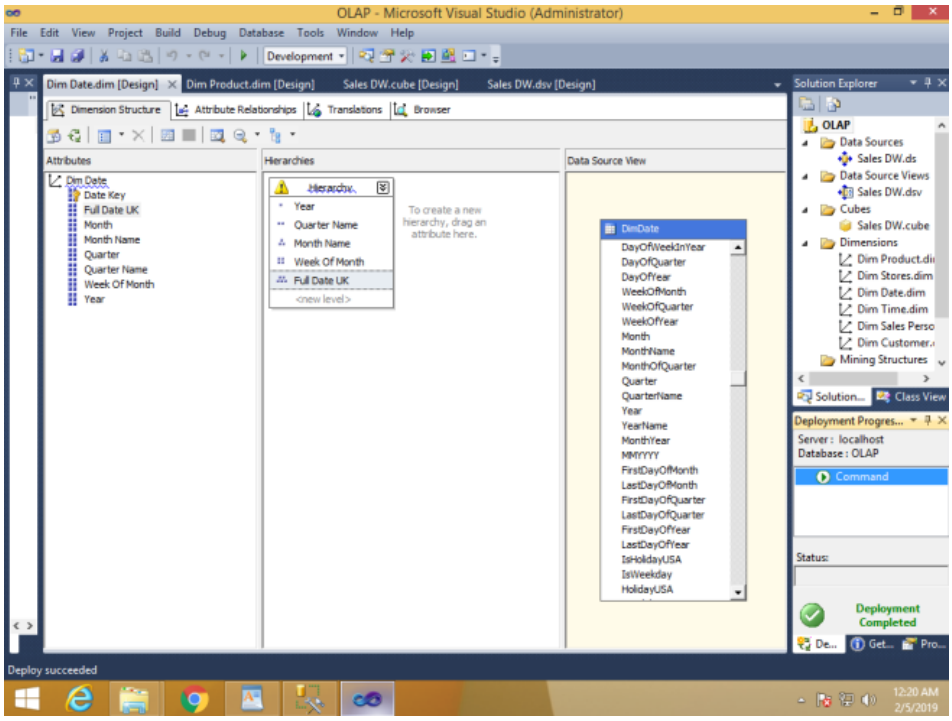
Do following changes and click on Apply & ok



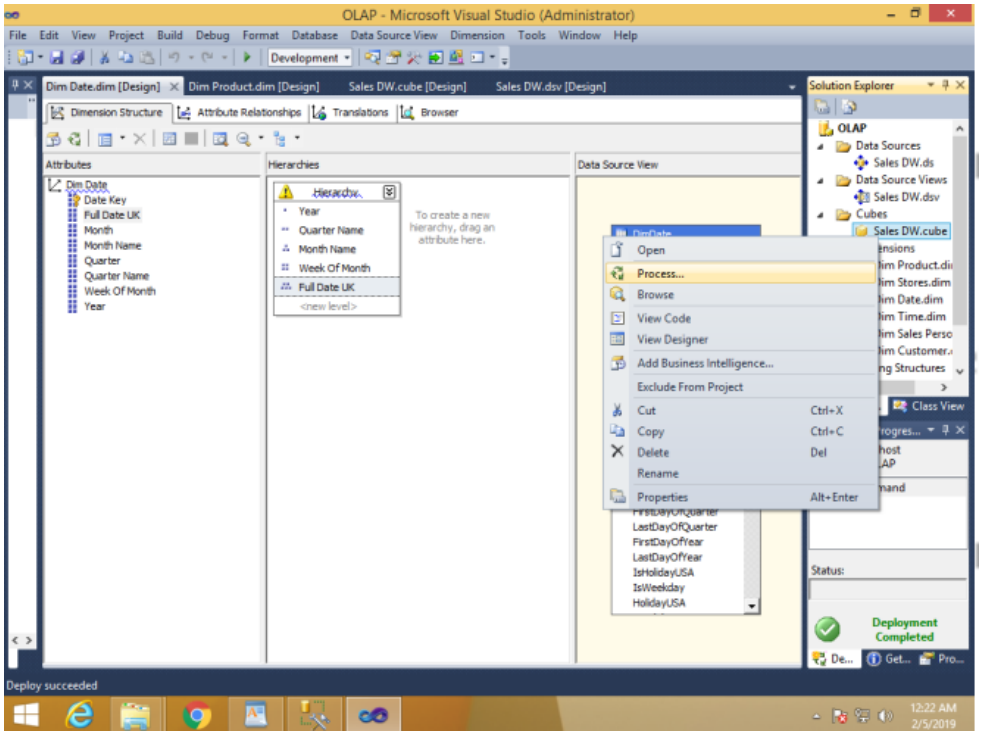
Right click on project name → Deploy



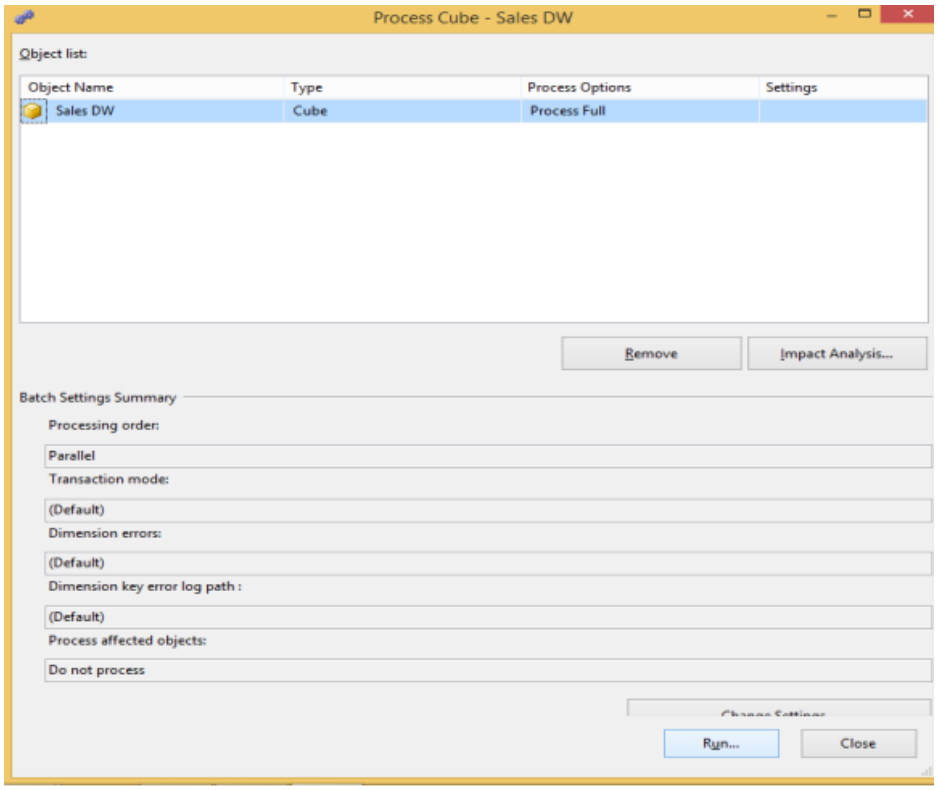
Deployment successful

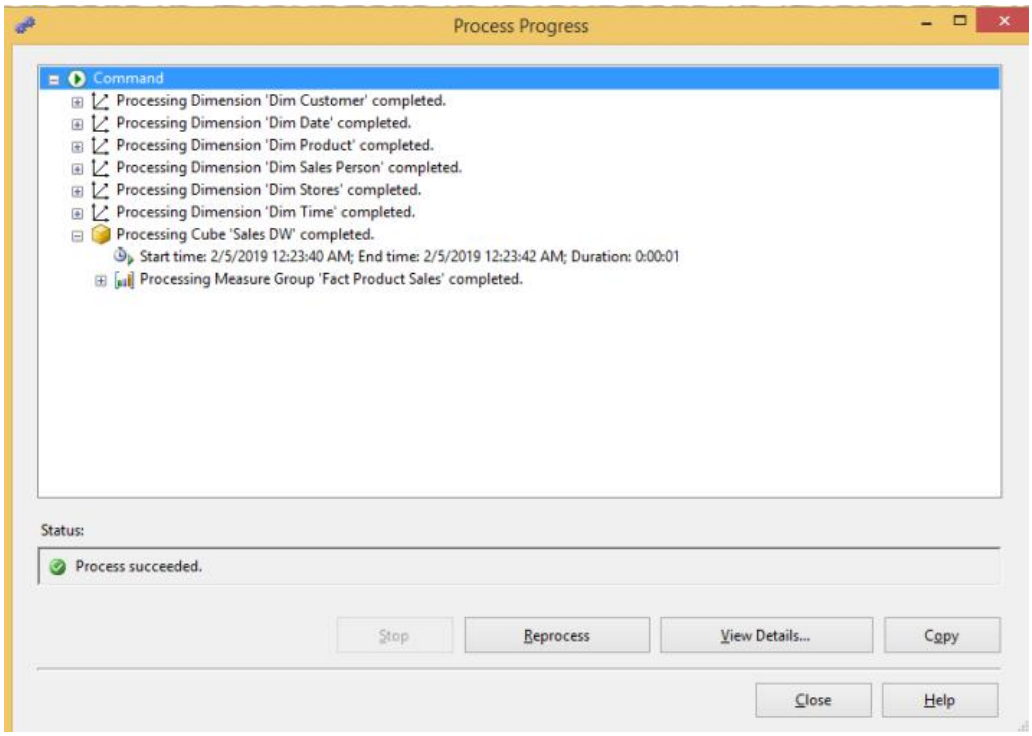


To process cube right click on Sales_DW.cube → Process

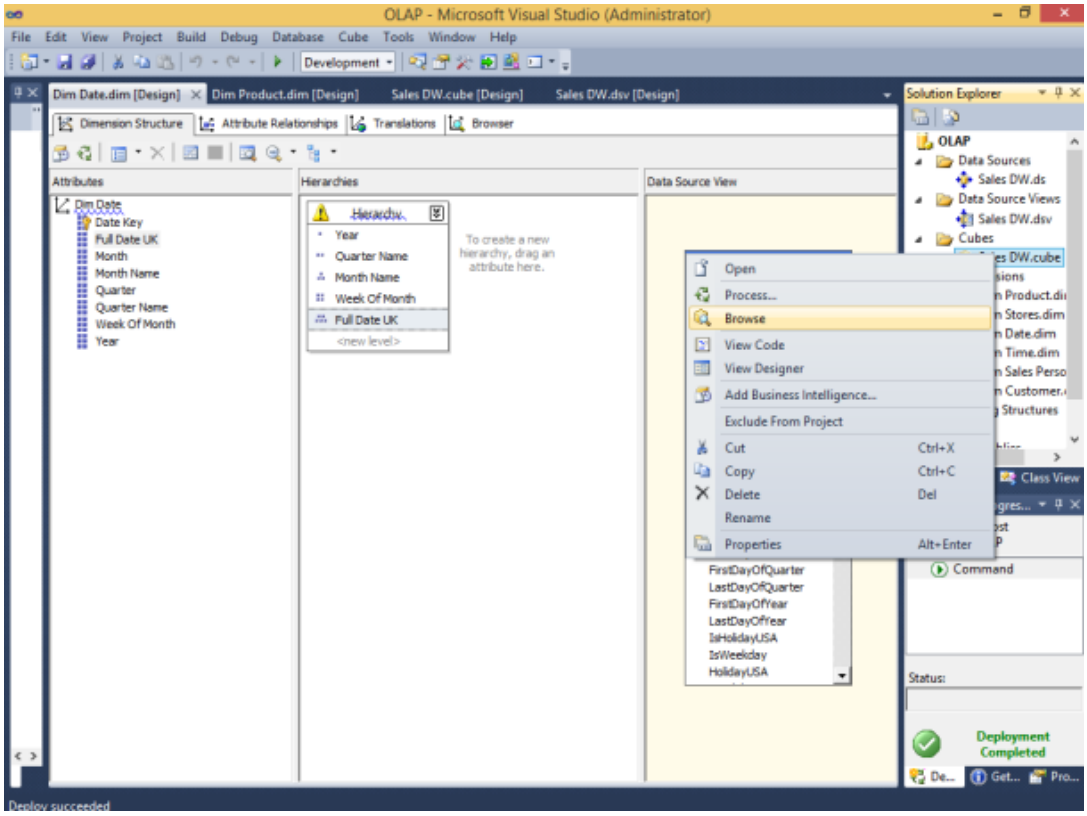


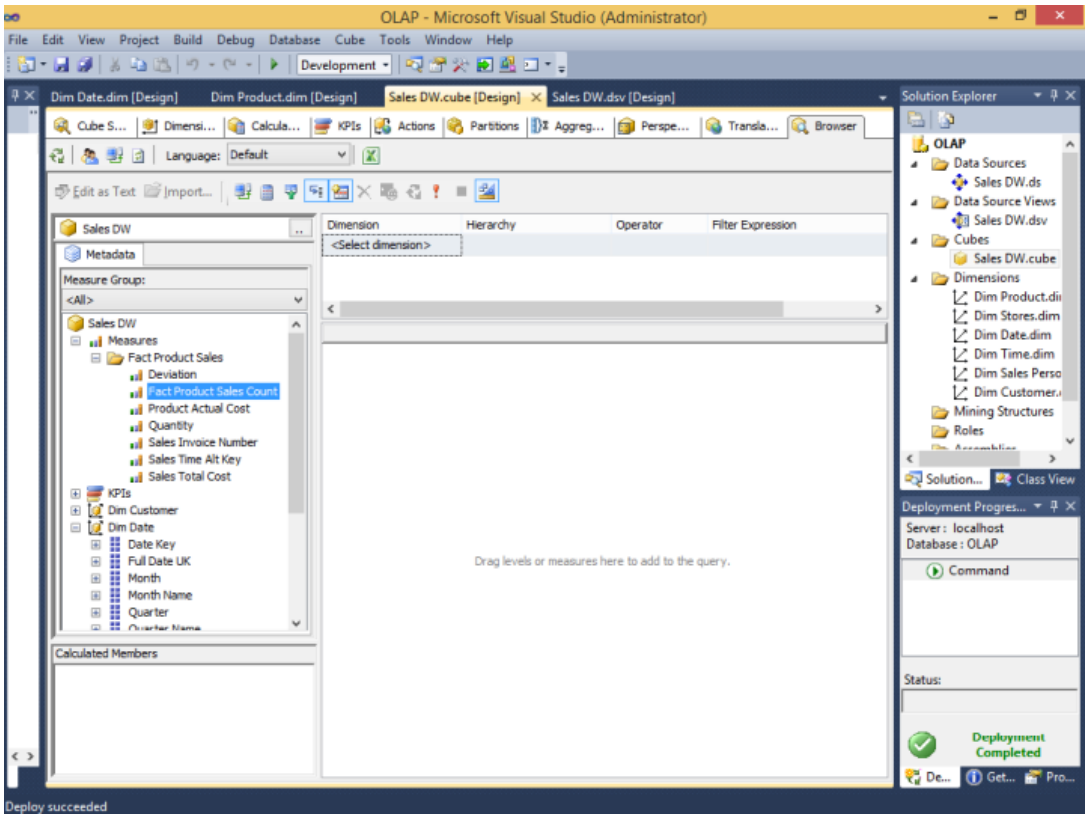
Click run





Browse the cube for analysis in solution explorer


OLAP - Microsoft Visual Studio (Administrator)
File Edit View Project Build Debug Database Cube Tools Window Help
Development
Dim Date.dim [Design]
Dim Product.dim [Design]
Sales DW.cube [Design]
Sales DW.dsv [Design]
Dimension Structure
Attribute Relationships
Translations
Browser
Attributes
Dim Date
Date Key
Full Date UK
Month
Month Name
Quarter
Quarter Name
Week Of Month
Year
Hierarchies
Hierarchy
Year
Quarter Name
Month Name
Week Of Month
Full Date UK
<new level>
To create a new hierarchy, drag an attribute here.
Data Source View
Open
Process...
Browse
View Code
View Designer
Add Business Intelligence...
Exclude From Project
Cut Ctrl+X
Copy Ctrl+C
Delete Del
Rename
Properties Alt+Enter
FirstDayOfQuarter
LastDayOfQuarter
FirstDayOfYear
LastDayOfYear
IsHolidayUSA
IsWeekday
HolidayUSA
Solution Explorer
OLAP
Data Sources
Sales DW.ds
Data Source Views
Sales DW.dsv
Cubes
Class View
Command
Status:
Deployment Completed
Deploy succeeded


OLAP - Microsoft Visual Studio (Administrator)
File Edit View Project Build Debug Database Cube Tools Window Help
Development
Dim Date.dim [Design]
Dim Product.dim [Design]
Sales DW.cube [Design]
Sales DW.dsv [Design]
Cube S...
Dimensi...
Calcula...
KPIs
Actions
Partitions
Aggreg...
Perspe...
Transla...
Browser
Language: Default
Edit as Text
Import...
Sales DW
Metadata
Measure Group:
<All>
Sales DW
Measures
Fact Product Sales
Deviation
Fact Product Sales Count
Product Actual Cost
Quantity
Sales Invoice Number
Sales Time Alt Key
Sales Total Cost
KPIs
Dim Customer
Dim Date
Date Key
Full Date UK
Month
Month Name
Quarter
Calculated Members
Dimension
Hierarchy
Operator
Filter Expression
<Select dimension>
Drag levels or measures here to add to the query.
Solution Explorer
OLAP
Data Sources
Sales DW.ds
Data Source Views
Sales DW.dsv
Cubes
Sales DW.cube
Dimensions
Dim Product.dim
Dim Stores.dim
Dim Date.dim
Dim Time.dim
Dim Sales Perso
Dim Customer.
Mining Structures
Roles
Solution...
Class View
Deployment Progres...
Server : localhost
Database : OLAP
Command
Status:
Deployment Completed
Deploy succeeded